Evaluation of Federated Learning against Centralized Learning for Multi-Label Thoracic Pathology Classification

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The adoption of electronic health records (EHRs) has increased substantially over the last few decades, driven by technological permeation and legislative mandates like the HITECH Act of 2009. The HITECH Act effectively authorized the Centers for Medicare and Medicaid Services (CMS) to offer monetary incentives to eligible healthcare providers to encourage the adoption of EHR use (Basil et al., 2022). Within just five years, 74% of office-based physicians and 96% of hospitals in the U.S. had certified EHR systems in place (Basil et al., 2022). Nevertheless, the digitization of health records has resulted in escalating health data privacy concerns, as the migration to electronic format has rendered sensitive patient data a high-value target for cybercriminals to exploit. Privacy concerns are further exacerbated by centralized EHR systems, where health records are stored in a single location and managed by a central authority. These frameworks elevate the risk of exposing sensitive data, as centralizing the storage of healthcare data means that a successful breach could compromise the sensitive records of millions of patients. Furthermore, the high concentration of sensitive data in a single location makes centralized systems vulnerable to cybercriminal exploitation. Additionally, centralized EHR systems face difficulties in being compliant with data privacy regulations such as HIPAA. In light of these risks, a decentralized approach provides a more reliable structure for protecting sensitive patient information and adhering to privacy regulations.

Federated Learning (FL) leverages these decentralized systems by training machine learning models across local servers and transmitting only gradients to a central server to update a global model. While FL enables multi-party collaboration without disclosing raw data, it faces challenges including communication overhead, potential data reconstruction from gradients, and performance instability due to non-IID data distributions. This creates a fundamental trade-off between maximizing model performance and preserving privacy. To evaluate the impact of this tradeoff, we compare the performance of an FL approach against a traditional centralized model where the entire dataset is pooled together in a single server. To ground this research, we perform transfer learning with DenseNet121, a pretrained convolutional neural network (CNN), and apply it to the National Institutes of Health (NIH) Chest X-Rays Dataset for a multi-label classification task involving fourteen thoracic diseases. This dataset models real-life settings where multi-site medical records are typically large-scale and siloed due to strict privacy constraints, making this challenge inherently a big data problem.

Materials and Methods

Dataset

The dataset used is based on the National Institutes of Health (NIH) Chest X-Ray dataset. This dataset originates from the Picture Archiving and Communication System (PACS) database at the NIH Clinical Center. It is comprised of 112,120 frontal-view X-ray images from 30,805 unique patients (Wang et al., 2017). As shown in Table 1, the dataset includes fourteen common thoracic diseases. The disease labels were mined using natural language processing (NLP) techniques.

Table 1: *Thoracic Disease Labels in the NIH Chest X-Ray Dataset*

Labels
Atelectasis
Consolidation
Infiltration
Pneumothorax
Edema
Emphysema
Fibrosis
Effusion
Pneumonia
Pleural Thickening
Cardiomegaly
Nodule
Mass
Hernia

Note. Adapted from Wang et al. (2017).

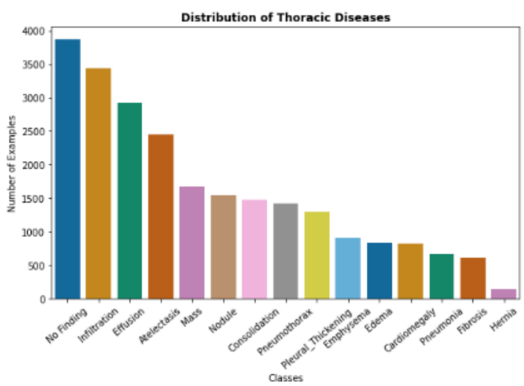
Our adapted version of the NIH Chest X-Ray dataset models a federated scenario with non-IID characteristics across three hospital silos and an additional out-of-distribution test set (Exalsius, n.d.). The dataset was partitioned using a scoring algorithm to model real-world data heterogeneity. As such, each hospital silo has different patient populations, imaging conditions, and disease distributions. Within each hospital, an 80/10/10 split was performed, and to prevent data leakage, splits within each hospital are patient-disjoint. Additionally, each patient appears in only one hospital silo. We downsampled the dataset, reducing its size from 45.1GB to approximately 9.02GB, representing ~20% of the original dataset.

Data Exploration

While FL preserves data locality by keeping each hospital's data siloed within its own server, the centralized setting aggregates the data from all hospitals into a single pooled dataset. Figure 1 shows the class distributions for the pooled dataset used by the centralized baseline.

Figure 1

Centralized Dataset Class Distribution

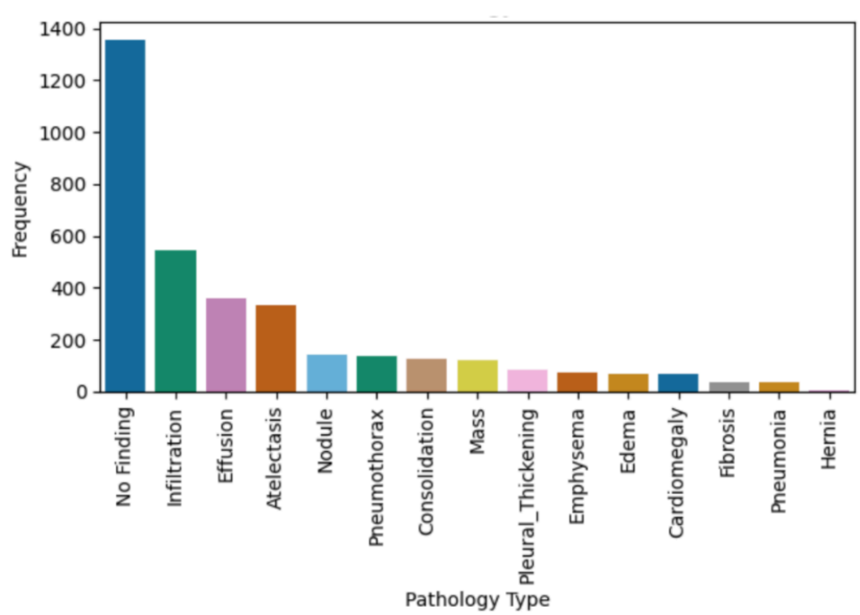


Pathology Label	Frequency	Percentage (%)
No Finding	3867	30.79
Infiltration	3433	27.33
Effusion	2919	23.24
Atelectasis	2456	19.56
Mass	1672	13.31
Nodule	1552	12.36
Consolidation	1471	11.71
Pneumothorax	1422	11.32
Pleural_Thickening	1295	10.31
Emphysema	913	7.27
Edema	838	6.67
Cardiomegaly	821	6.54
Pneumonia	664	5.29
Fibrosis	616	4.90
Hernia	144	1.15

As can be seen, the dataset exhibits class imbalance. This imbalance is further amplified in the siloed datasets. Figure 2 shows that Hospital A's data displays severe data imbalance as a "long-tail" distribution. Common thorax diseases like infiltration significantly outnumber rare conditions like hernias. This imbalance is further exacerbated by the multi-label nature of the task, where co-occurrence patterns vary across the hospital silos.

Figure 2

Hospital A Thoracic Pathology Distribution



Preprocessing

With regards to preprocessing, we accounted for high-frequency noise and low local contrast. Beyond standard resizing from 1024 x 1024 to 224 x 224 pixels, we applied Contrast-Limited Adaptive Histogram Equalization (CLAHE). This technique is commonly used for chest X-rays as it enhances the visibility of subtle features like small pulmonary nodules or faint infiltrates without over-amplifying noise in uniform regions.

Asymmetric Loss (ASL)

The primary challenge in this multi-label classification task is the highly skewed distribution of positive and negative samples per class. Most images contain only a small fraction of the fourteen possible labels, making the number of positive samples per class significantly lower than the number of negative samples. This imbalance is problematic, as it can lead to under-emphasizing gradients from positive labels during the optimization process, resulting in poor accuracy (Ridnik et al., 2021). In an attempt to mitigate this, we moved beyond Binary Cross-Entropy (BCE) and implemented Asymmetric Loss, Equation (1), as proposed by Ridnik et al. (2021).

$$\mathrm{ASL} = \begin{cases} L_+ = (1-p)^{\gamma_+} \log(p), \\ L_- = (p^m)^{\gamma_-} \log(1-p_m) \end{cases} \quad \text{where} \quad p_m = \max(p-m, 0) \tag{1}$$

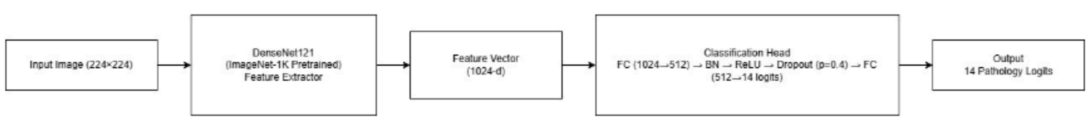
ASL is based on two key properties: soft down-weighting and hard thresholding. The former reduces the contribution of negative samples to the loss when their probability is low using asymmetric focusing, meaning that easy negatives are down-weighted during training. The latter goes a step further and completely eliminates the contribution from extremely easy negative samples through the use of shifted probability p_m, which discards samples whose predicted probability p falls below a margin m. Soft thresholding is achieved through focusing parameters γ_+ and γ_-, while hard thresholding is achieved via the probability margin m. Using ASL allows the model to focus on difficult samples and rare positive cases without being biased by the high prevalence of “No Finding” or common pathologies.

Model Architecture

Due to its consistently strong performance reported in prior works such as Irvin et al. (2019) and Gozes and Greenspan (2019), the DenseNet121 architecture was chosen for this experiment. DenseNet121 is a specific instance of DenseNet, a convolutional neural network that connects each layer to every other layer in a feed-forward manner. We use the pretrained DenseNet121 provided by the PyTorch torchvision library, which was trained on the ImageNet-1K dataset. As shown in Figure 3, we replaced the original classification layer with a custom multi-layer classification head designed to handle the fourteen pathology classes.

Figure 3

DenseNet121-Based Classification Architecture With Custom Head



Note. The model consists of a pretrained DenseNet121 backbone followed by a custom classification head.

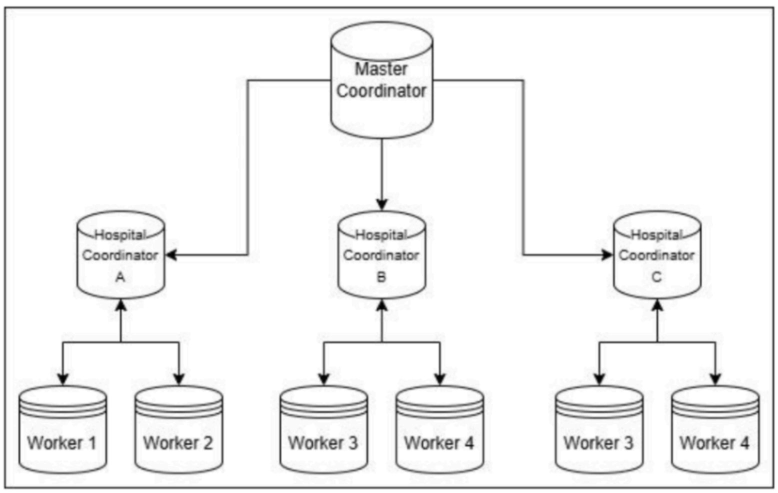
The classification head takes the 1024-dimensional features from DenseNet121 and projects them into a 512-dimensional space using a fully connected layer. Batch normalization is then employed to stabilize training and allow for higher learning rates. This is followed by a ReLU activation layer to introduce non-linearity, a dropout layer with rate $p = 0.4$ to prevent overfitting, and a final fully connected layer that maps the 512-dimensional hidden units to fourteen output logits corresponding to the multi-label thoracic pathology predictions.

Centralized Baseline Model. The baseline model was trained in a centralized fashion, pooling each split's data from the three hospitals. We used the AdamW optimizer with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-3}. A key component of our methodology is dynamic threshold tuning. Because medical findings vary in their optimal detection probability, we do not use a fixed 0.5 threshold. Instead, at the end of each validation epoch, we calculate the threshold that maximizes the F1-score for each of the fourteen classes independently. These optimal thresholds are then registered as model buffers and applied during inference on both the In-Distribution (ID) and Out-of-Distribution (OOD) test sets to ensure a fair comparison of model generalization.

FL Framework. To simulate a multi-institutional collaborative environment, we implemented an FL framework using the Flower library. We employed a 1-3 architecture, consisting of a single central server and three hospital clients representing Hospitals A, B, and C. The coordination process followed a synchronous Federated Averaging (FedAvg) strategy, customized to accommodate the multi-label nature of the task. In each communication round, the server broadcasts the current global model parameters to all clients. Each client then performs local training for one epoch using the AdamW optimizer and siloed hospital data. Upon completion of local training, clients transmit updated weights back to the server. To ensure the global model remains representative of the total data distribution, we implemented a weighted metric aggregation function. This function calculates the macro-AUROC, F1-score, and ASL by weighting each client's contribution based on their local sample size.

Hierarchical Federated Learning (HFL). For comparison to the federated architecture, we developed an HFL setup (Figure 4). It follows a 1-3-6 architecture: one master coordinator, three hospital coordinators, and six Distributed Data Parallel (DDP) workers (two workers per hospital silo). The inter-hospital communication between the coordinators is managed via Flower. The intra-hospital training is managed by DDP, where each hospital coordinator acts as a DDP master for its local workers. When the hospital coordinator receives global weights from the master, it orchestrates a local DDP training session. The local workers synchronize gradients via an NCCL backend on a dedicated local port, effectively treating the hospital's multi-node cluster as a single entity.

Figure 4
Hierarchical Federated Learning Architecture



Evaluation Metrics

A critical component of our study is to measure the fundamental trade-off between maximizing model performance and preserving privacy. As such, to evaluate the performance of the FL approach and the traditional centralized baseline, we focused on three categories of metrics: performance-based, efficiency-based, and privacy-based measures. Although FL preserves privacy through data siloing, we further evaluate privacy leakage in both the FL and centralized settings by using Membership Inference Attacks (MIA).

Performance Metrics. Consistent with prior multi-label classification studies, we evaluate performance using Area Under Curve Receiver Operating Characteristic (AUROC), precision, recall, and F1-score. To measure the robustness of the baseline, we defined the generalization gap as: $|AUROC_{ID} - AUROC_{OOD}|$. This metric quantifies the performance degradation when the model encounters data from Hospital D, which features distinct patient demographics not seen during training.

AUROC. The ROC curve evaluates the model's ability to distinguish between positive and negative classes across different decision thresholds, while AUROC summarizes the model's overall class separability. A higher AUROC means the model is better at ranking true positive cases above negative ones across all thresholds, while an AUROC of 0.5 means that there is a 50% chance that a random positive sample is ranked higher than a randomly chosen negative example, indicating that the model is no better than random guessing. We compute this metric on a per-class basis by treating each label as an independent binary classification task and additionally report macro AUROC.

Precision. This metric measures the accuracy of positive predictions as shown in (2):

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

In a medical setting, precision is important as false positives (i.e., cases where a healthy patient is labeled as sick) cause unnecessary stress and expenses for patients. Lowering false positives also has the additional benefit of ensuring that medical resources are not wasted on healthy individuals.

Recall. Also referred to as Sensitivity or True Positive Rate (TPR), recall measures how many positive cases the model successfully detects and is defined in Equation (3) as:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

Recall is highly important in a medical context because it may be life-threatening to miss a patient's condition.

F1-Score. In practice, there exists a trade-off between precision and recall. In order to provide a balanced view of both metrics, the F1-score is commonly used. As shown in (4), this score is the weighted harmonic mean of precision and recall scores, providing a single value that balances both measures.

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Efficiency and Resource Overhead. When evaluating FL, computational efficiency and communication costs are key factors to consider. This is because if the overhead associated with model updates and synchronization is too high, then it can undermine the privacy benefits that FL is designed to provide. As such, we implemented a custom metrics logger to capture metadata for every round. We measured communication overhead, tracking the total bytes transferred per client to calculate bandwidth efficiency, defined as the improvement in AUROC per megabyte of data exchanged. Additionally, we measure full-round latency, accounting for both the client compute time and the server's aggregation overhead.

Privacy Evaluation Metrics. As stated at the outset, sensitive patient data is a high-value target for adversaries to exploit. Such data combines personal identifiers, financial information, and the health history of a patient, making it especially attractive for identity theft, fraud, and blackmail. In the context of machine learning, adversaries may employ MIA techniques in order to determine, with some level of confidence, if a specific individual's data was used to train a model. This inference can enable cybercriminals to learn sensitive information about patients, such as their medical history and association with a specific healthcare provider. A successful attack also constitutes a violation of privacy regulations. As Yeom et al. (2018) explain, the effectiveness of MIA is highly correlated with model overfitting. Overfitted models tend to partially memorize specific details about training examples, leading them to produce outputs with distinctive patterns compared to the outputs produced from unseen data. For instance, a model may exhibit higher confidence scores with training samples than with unseen data, as it has already been exposed to the former during training, so the model is more confident about its prediction.

Threshold-Based MIA. To evaluate this vulnerability, we implement a threshold-based MIA approach inspired by Yeom et al. (2018). Threshold-based MIA relies on the fact that there is a generalization gap between the train and test splits, so membership becomes a statistical signal. Yeom et al. (2018) show that membership inference can be performed by comparing a sample's loss to a threshold derived from the model's training loss. While samples below the threshold are inferred as members, samples above the threshold are inferred as non-members. We extend this idea by incorporating threshold-free metrics based on ROC analysis. Our approach works by computing the loss scores of members and non-members. We then assign labels to the samples—1 for members and 0 for non-members—resulting in a set of scores and labels from all samples. The ROC curve and AUC values are then computed using the scores and labels to evaluate how well the scores separate members and non-members across all possible thresholds. To compute balanced accuracy, we select the optimal threshold using Youden's J statistic as shown in (5), (6), and (7):

$$J = \mathrm{TPR} - \mathrm{FPR} \tag{5}$$

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{6}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} \tag{7}$$

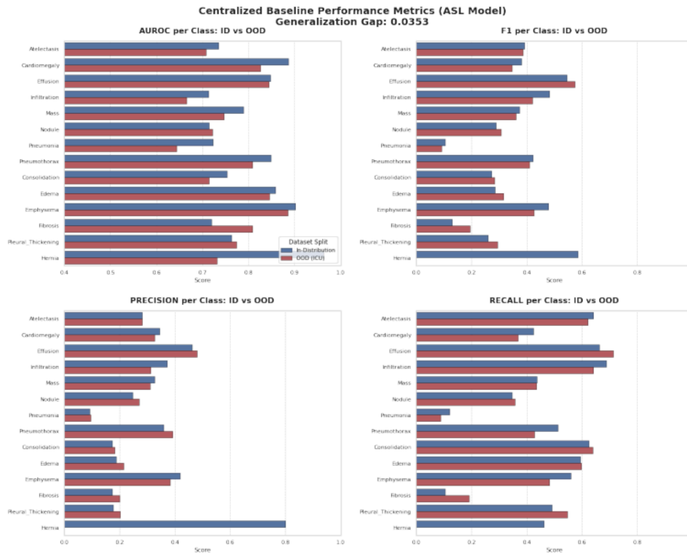
This threshold is used to generate binary predictions to evaluate the performance at the point that best balances the true positive and false positive rates.

Results and Evaluation

Centralized Baseline Evaluation and Performance

The centralized baseline was evaluated over 30 epochs. The results are shown in Figure 5. Per-class AUROC values range from 0.60 to 0.90, indicating moderate to strong discriminative performance, but with performance noticeably varying between pathologies. Precision stands around 0.20 for most classes and 0.80 for hernia. This indicates that the model is over-predicting the positive class for most pathologies, although for hernia, it is correct 80% of the time. As for recall, most classes have a value of approximately 0.40, suggesting that the model is failing to identify many true positive cases, leading to high false negative rates. These results are summarized in the F1-score graph, where values stand around 0.40. In regards to ID and OOD test set performance, it can be seen that for the most part, the model performs similarly on both test sets, but with slightly higher performance on the ID test set. This is expected, as the ID test data is drawn from the same distribution as the training set, while the OOD test set exhibits a slightly different distribution due to being sourced from Hospital D.

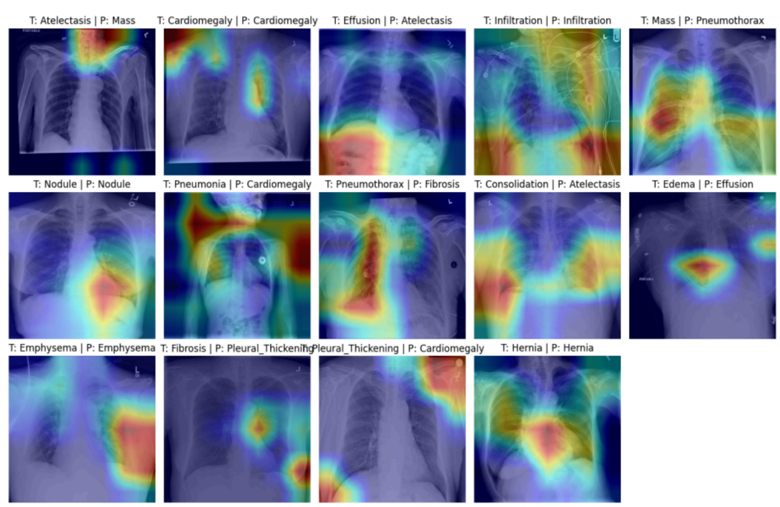
Figure 5



The Grad-CAM visualizations shown in Figure 6 illustrate the active regions of the feature maps that the model looks at when making its predictions for each class. As can be seen, the model was able to accurately predict five thoracic diseases correctly, namely cardiomegaly, infiltration, nodule, emphysema, and hernia. However, without expert annotation, it is difficult to determine whether the highlighted regions capture clinically relevant features.

Figure 6

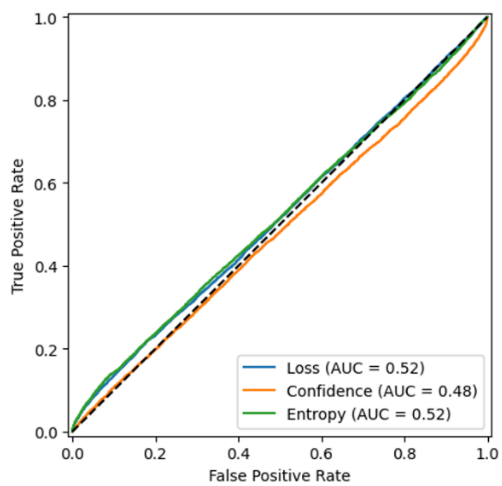
Grad-CAM Heatmaps for the Centralized Thoracic Pathology Model



With regards to the MIA performance of the centralized model, as can be seen in Figure 7, the AUROC is close to 0.5, which suggests that the attack is equivalent to random guessing, meaning that the adversary would not be able to tell whether a patient's data was among the training samples. Confidence and entropy were also calculated in order to see if different results emerged, but as can be seen, the AUROC values for these metrics also correspond to random guessing.

Figure 7

Centralized Model Membership Inference Attack ROC Curve



The results are further summarized in Table 2, which additionally reports balanced accuracy. The balanced accuracy for all methods is around 50%, further reinforcing the claim that the attacker can do no better than randomly guess.

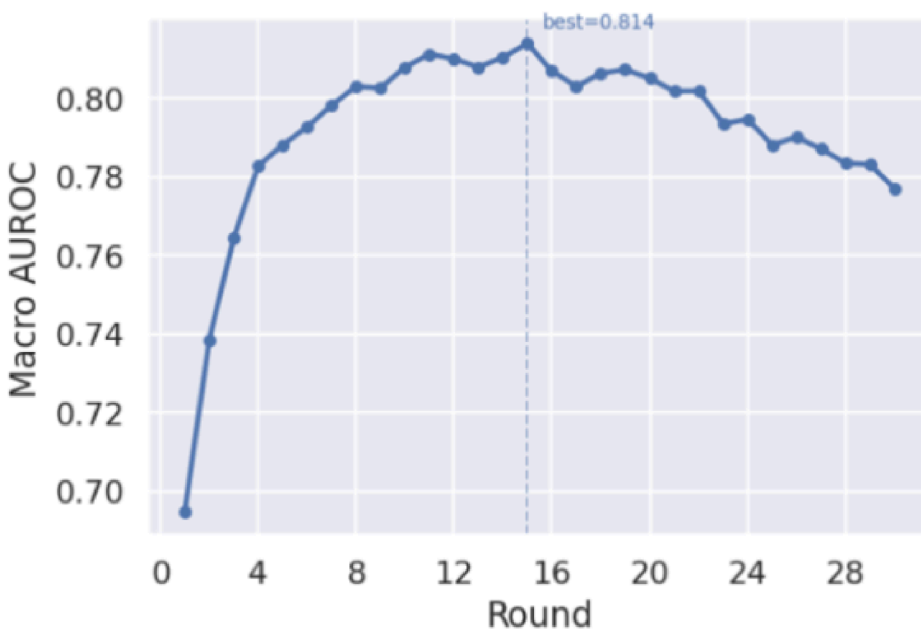
Table 2: *Centralized MIA Performance (AUROC and Balanced Accuracy)*

Method	AUROC	Balanced Accuracy
Loss	0.52	0.52
Confidence	0.48	0.50
Entropy	0.52	0.52

Federated Evaluation and Performance

The federated global model was evaluated over 30 communication rounds. It demonstrated a consistent upward trajectory in utility, with the macro-AUROC plateauing at approximately 0.78 by round 30, as shown in Figure 8. This convergence suggests that the customized FedAvg strategy effectively aggregates the non-IID features present in the heterogeneous hospital datasets without requiring raw data centralization.

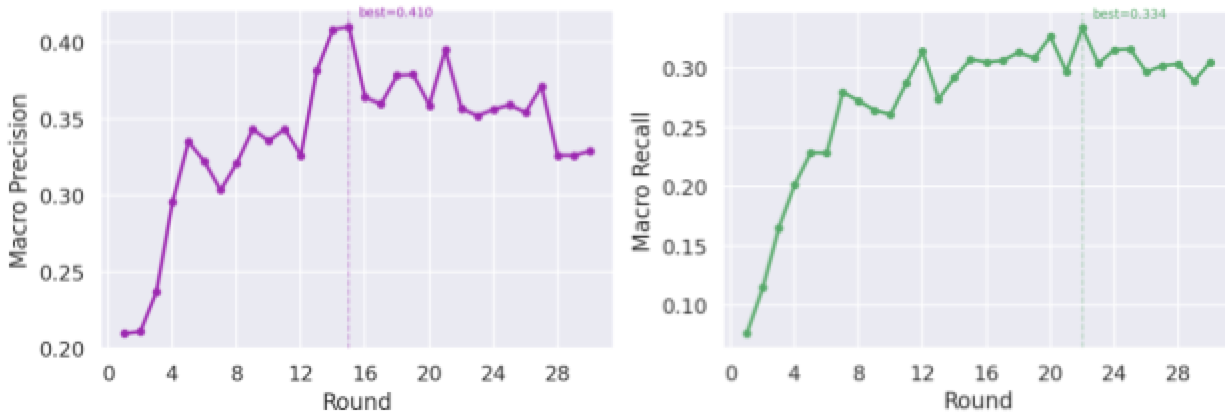
Figure 8
Macro-AUROC over Training Period for Federated Model



Both macro-precision and macro-recall (Figure 9) plateaued alongside AUROC, indicating that the dynamic threshold tuning implemented during validation was successful in balancing the precision-recall tradeoff across the fourteen pathologies. However, the aggregated validation loss showed a sharp initial decline, followed by steep increases after round 16. This is an indication of overfitting, which was not remedied by a learning rate scheduler or more training data.

Figure 9

Federated learning macro-precision and macro-recall over the training duration



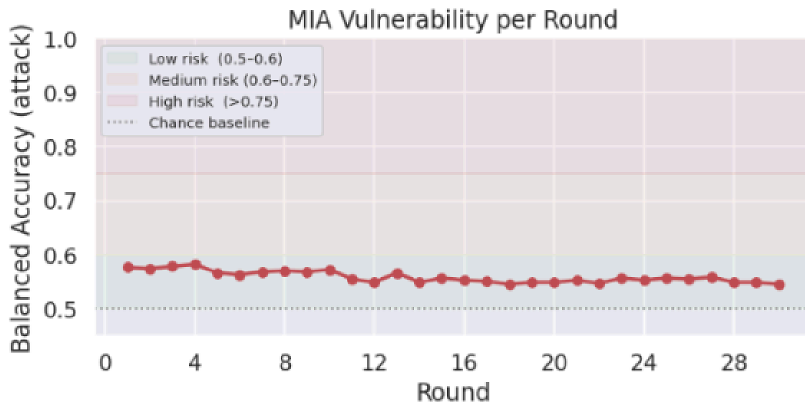
A primary objective was to quantify the performance degradation when moving from ID data to OOD data (Table 3). The results indicated a generalization gap of ~0.0526 in AUROC. Analysis of individual pathologies reveals that the model maintains high robustness for conditions with clear anatomical markers, such as cardiomegaly and pleural effusion. Conversely, texture-heavy conditions like pneumonia and infiltration exhibited the most significant performance drops, likely due to variations in imaging hardware and the type of imaging used (X-ray).

Table 3: *Performance Under In-Distribution and Out-of-Distribution Settings*

Metric	**In-Distribution (A, B, C)**	**OOD (D)**	**Generalization Gap (Δ)**
Macro-AUROC	0.81	0.76	0.05
Macro-F1	0.42	0.38	0.04

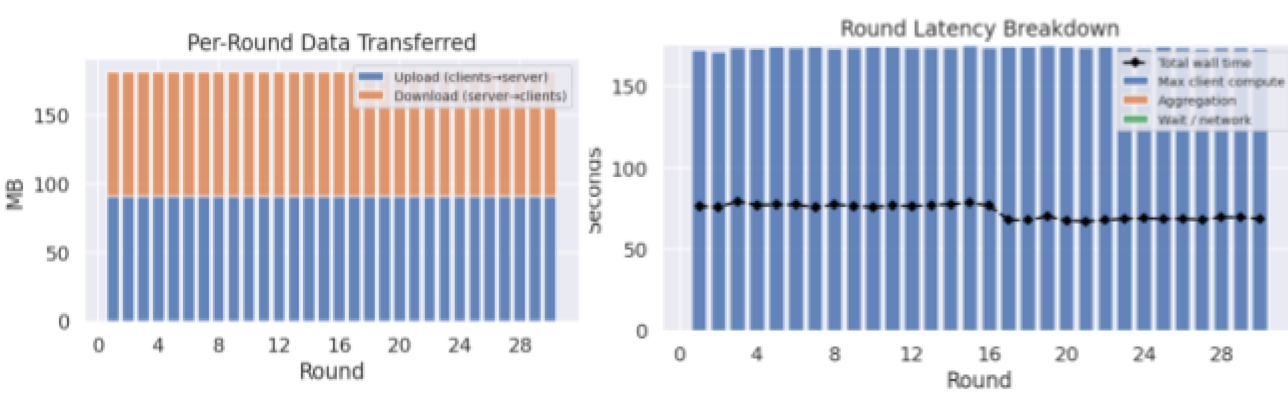
We monitored the MIA vulnerability score to assess the privacy-utility trade-off during training (Figure 10). We see a consistent downward trend in MIA vulnerability as the rounds progress. It peaks at ~0.582 during round 4, and reaches its lowest of ~0.545 during round 30. It may have continued downwards with more training, but this score is already a very good indication of privacy being maintained.

Figure 10
MIA Vulnerability per Round



Finally, with regards to communication overhead, the FL architecture is relatively lean. Each round, ~80MB was uploaded to the coordinator and ~90MB was downloaded from the coordinator to each worker. Undoubtedly, most of the time spent each round was during the client computation phase (Figure 11). The network communication and aggregation times were exceedingly minuscule in relation to compute time, so the gains from parallelization far outweigh any overhead cost incurred from the architecture.

Figure 11
Communication overhead for federated learning setup over training period



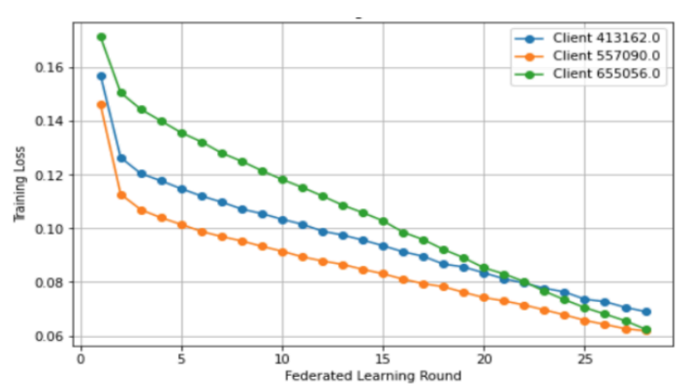
HFL Evaluation and Performance

For the federated model, the global performance improved during the early communication rounds. The AUROC curve increased rapidly in the first few rounds and then stabilized around the middle of training. This indicates that the global model successfully learned useful visual features from the distributed hospital clients. At the same time, the validation loss initially decreased, suggesting better generalization. However, after approximately 10–15 rounds, validation loss began to increase while AUROC remained mostly stable or slightly decreased. This suggests that longer training did not continue improving generalization and may have introduced mild overfitting. Based on this behavior, the best model checkpoint likely occurs around the point where validation loss is lowest and AUROC is near its peak, rather than at the final communication round.

The per-client training loss curves in Figure 12 show that all three hospital clients improved consistently across federated rounds. Each client's loss decreased steadily, which confirms that local training was successful at each hospital. However, the clients did not begin with the same loss values and did not decrease at exactly the same rate. This is expected because the dataset was intentionally partitioned into non-IID hospital silos.

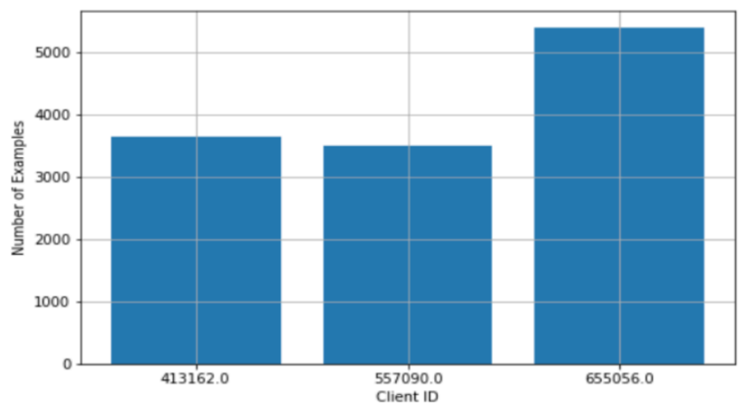
Figure 12

Per-Client Training Loss Across Rounds



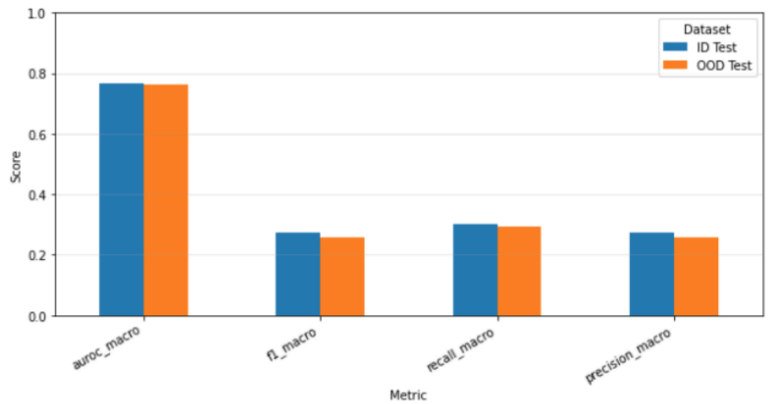
The client dataset-size graph (Figure 13) shows that the clients had different numbers of examples, with one hospital containing more samples than the others. This difference affects each client's contribution to the global model and supports the use of weighted metric aggregation based on local sample size. Without weighted aggregation, smaller and larger hospitals would contribute equally, which could make the global model less representative of the overall data distribution.

Figure 13
Dataset Size Per Client



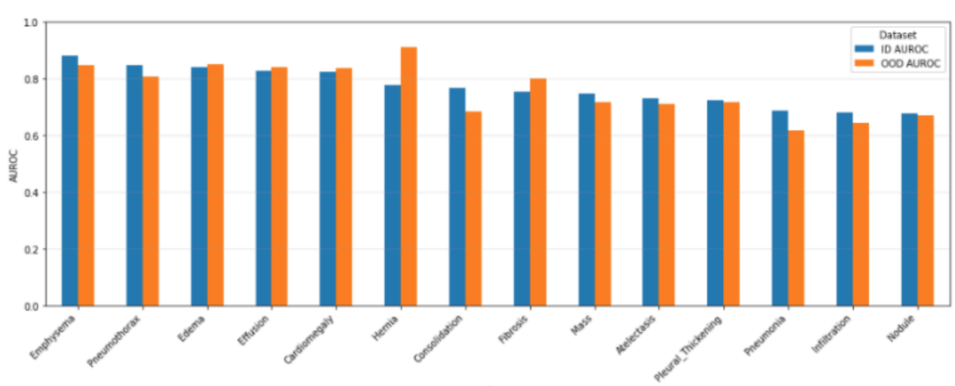
The ID vs. OOD comparison provides an important measure of model robustness. Figure 14 shows that the model performed similarly on the in-distribution and out-of-distribution test sets, with only a small drop in macro AUROC, F1-score, recall, and precision. This small generalization gap suggests that the learned representation was not limited only to the training hospital distributions.

Figure 14
Overall Performance: ID vs. 00D



The per-class AUROC comparison in Figure 15 further shows that some pathologies generalized better than others. Several classes maintained similar AUROC across ID and OOD settings, while others showed a noticeable drop. This indicates that rare or visually ambiguous pathologies remain more difficult to generalize. This result is important because it shows that the model's overall macro AUROC does not fully explain class-level behavior. In medical classification, evaluating each pathology separately is necessary because poor performance on a rare disease may be hidden by stronger performance on more common diseases.

Figure 15
Per-Class AUROC: ID vs. OOD



Overall, the results show that federated learning was successful in training a multi-label thoracic pathology classifier across distributed hospital clients. The model achieved stable AUROC performance, decreasing client training loss, and reasonable ID-to-OOD generalization. However, the increasing validation loss in later rounds suggests that early stopping and careful checkpoint selection are necessary. The experiments also show that non-IID hospital data and class imbalance remain major challenges, making macro-level evaluation, client-level analysis, and OOD testing essential for understanding the true performance of the system.

Comparative Analysis & Conclusion

The three architectures present a clear trade-off between predictive utility, data privacy, and infrastructure scalability. From the experiments above, it can be observed that the centralized and federated approaches achieve comparable performance across the given evaluation metrics. This indicates that although the centralized baseline has access to the fully pooled dataset, it does not translate to meaningful performance gains. However, this lack of performance gains may also be due to all models overfitting to the training data, which naturally limits generalization across all the approaches and could potentially be masking any potential benefits that a pooled dataset may provide. Nevertheless, pooling the data together comes at a significant privacy cost. The requirement to store over 112,000 patient X-rays in a single location creates a high risk for data breaches and poses substantial hurdles to patient confidentiality.

The simple federated model successfully maintains data locality, ensuring that raw patient pixels never leave the hospital coordinator. The global model achieved a competitive AUROC of 0.78, nearing the performance of the centralized baseline. Analysis of the validation loss suggests that while the model converges quickly, utility begins to plateau after round 20. This indicates that extended training rounds may lead to memorization of local silo features, necessitating early stopping to preserve both generalization and privacy.

The HFL model represents the most operationally robust approach for real-world medical environments. By introducing a second tier of distribution, where each hospital coordinator manages a DDP group, the system can scale to handle significantly larger local datasets without a linear increase in per-round latency. While the HFL offers the best path for high-throughput training in multi-node hospital environments, it introduces additional complexity in network orchestration, specifically regarding port management and synchronization across coordinators.

In summary, the centralized model did not demonstrate performance gains, and it is increasingly impractical in the face of modern privacy regulations. Standard FL offers a viable alternative for smaller-scale collaborations, but the HFL approach stands out as the superior choice for real hospital networks. It provides the necessary infrastructure to handle the massive data volumes typical of modern clinical imaging departments while maintaining privacy guarantees inherent in federated systems.

Contributions

Jason Curcio	• Centralized baseline implementation & training • FL model implementation & training • HFL model implementation & training
Sofia Catalan	• Centralized baseline implementation & training • Exploration • Centralized MIA
Chirag Prasad Deshpande	• HFL Implementation and MIA implementation on HFL • Tested out how to model works for OOD vs IID

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